

CLOUD COMPUTING

Complete Study Notes — Day 1, Day 2 & Day 3

Topics Covered

Chapter 1 — What is Cloud?	Chapter 2 — Public vs Private Cloud
Chapter 3 — Virtualization	Chapter 4 — AWS vs Others
Chapter 5 — IAM in AWS	Chapter 6 — Practical: S3 + IAM Lab
Chapter 7 — EC2 in AWS	Chapter 8 — Practical: Jenkins on EC2
Chapter 9 — What is VPC + Components	Chapter 10 — How VPC Works + Why VPC



CHAPTER 9 — What is VPC & Its Components



Simple Explanation — The Gated Society

AWS is a huge apartment complex with thousands of flats.
By default everyone shares the same corridors — anyone can walk anywhere.

AWS gives you the option to build your OWN private gated society inside that complex — with your own walls, your own security guard, your own gate, and your own rules about who enters and exits.



That private gated society = VPC (Virtual Private Cloud)



Technical Definition



Definition:

VPC (Virtual Private Cloud) is a logically isolated section of the AWS Cloud where you can launch AWS resources like EC2, RDS, and Lambda in a virtual network that YOU define and control — with your own IP ranges, subnets, route tables, and gateways.

The Big Picture — VPC as a House

VPC = The land boundary (your private network space)

Subnets = Rooms inside the house

Internet Gateway = The main front door

Route Table = Signboards showing which road leads where

Security Group = Lock on each room's door

NACL = Security guard at each floor entrance

NAT Gateway = Back door — only for going out, not coming in

VPC Peering = A private tunnel to your neighbour's house

Component 1 — Subnets

PUBLIC SUBNET

Connected to Internet Gateway
Resources CAN be reached from internet
Used for: Web servers, Load Balancers
Example IP: 10.0.1.0/24

PRIVATE SUBNET

NOT connected to Internet Gateway
Resources CANNOT be reached from internet
Used for: Databases, App servers
Example IP: 10.0.2.0/24

Golden Rule: Web servers go in Public Subnet. Databases ALWAYS go in Private Subnet.
Analogy: Living room (public — guests can enter) vs Bedroom (private — family only)

Component 2 — Internet Gateway (IGW)

The Internet Gateway is the bridge between your VPC and the public internet.
It is AWS-managed, fully redundant — it never goes down.

Flow: Internet → Internet Gateway → Public Subnet → EC2 Web Server

Rules:

- One IGW per VPC — you can only attach one at a time
- Must be attached to VPC — just creating it is not enough
- Public Subnet only — private subnets cannot use it directly
- Two-way traffic — allows both incoming and outgoing internet traffic

Analogy: The MAIN FRONT DOOR of your gated society.

Component 3 — Route Table

A Route Table is a rulebook that tells traffic where to go inside your VPC.
Every subnet must be associated with a route table.

Route Table for PUBLIC Subnet:

- Destination: 10.0.0.0/16 → Target: Local (stay inside VPC)
- Destination: 0.0.0.0/0 → Target: Internet Gateway (go to internet)

Route Table for PRIVATE Subnet:

- Destination: 10.0.0.0/16 → Target: Local (stay inside VPC only)

(No internet route = completely private)

Analogy: Traffic SIGNBOARDS inside your gated society.

🔗 Component 4 — Security Groups

A Security Group is a virtual firewall at the INSTANCE level.
Every EC2, RDS, or Lambda inside your VPC gets a Security Group.

Example Rules (Web Server):

INBOUND: Port 80 (HTTP) Allow from 0.0.0.0/0 (anyone)
INBOUND: Port 443 (HTTPS) Allow from 0.0.0.0/0 (anyone)
INBOUND: Port 22 (SSH) Allow from MY IP only
OUTBOUND: All traffic Allow (default)

KEY: Security Groups are STATEFUL — if inbound Port 80 is allowed,
response traffic automatically goes out without a separate outbound rule.

Analogy: The LOCK on each individual room door inside the house.

🔗 Component 5 — NACL (Network Access Control List)

Feature	Security Group vs NACL
Level	Security Group = Instance level NACL = Subnet level
State	Security Group = Stateful NACL = Stateless
Rule Types	Security Group = Allow only NACL = Allow AND Deny
Rule Order	Security Group = All rules checked NACL = Number order
Default	Security Group = Deny all inbound NACL = Allow all
Think of as	Security Group = Room door lock NACL = Floor security guard

Traffic flow through both layers:

Internet → NACL (subnet level check) → Security Group (instance check) → EC2

Two layers = even if Security Group is misconfigured, NACL can still block threats.

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Component 6 — NAT Gateway

NAT (Network Address Translation) Gateway allows resources in a PRIVATE subnet to initiate outbound connections to the internet — but BLOCKS all unsolicited inbound connections from the internet.

Flow:

- Private Subnet (Database needs updates)
- ↓
- NAT Gateway (lives in PUBLIC Subnet)
- ↓
- Internet Gateway → Internet

Return traffic comes back the same path.
BUT no one from internet can INITIATE a connection to the database.

Analogy: A ONE-WAY MIRROR — you can see outside, nobody can see you.

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Component 7 — VPC Peering

[illegible]

VPC Peering is a private networking connection between TWO VPCs that allows them to communicate using private IP addresses — without going through the internet, a VPN, or any external gateway.

Example:

VPC A (Dev Team) 10.0.0.0/16

←──→
VPC Peering (private)

VPC B (Production) 172.16.0.0/16

Use case: Large companies with separate VPCs for dev, staging, production.
Teams talk privately without any data going over the public internet.

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Example:

VPC A (Dev Team) 10.0.0.0/16

The diagram consists of two parallel horizontal double-headed arrows. The top arrow is positioned below the text "VPC A (Dev Team)" and above the text "VPC B (Production)". The bottom arrow is positioned directly below the top one. Both arrows are centered horizontally relative to the space between the two VPC labels.

VPC Peering (private)

VPC B (Production) 172.16.0.0/16

Use case: Large companies with separate VPCs for dev, staging, production.
Teams talk privately without any data going over the public internet.

Default VPC vs Custom VPC

Feature	Default VPC	Custom VPC
Created by	AWS automatically	You create it from scratch
Available in	Every AWS account	Only if you build it
Use case	Quick testing, learning	Production, real apps
Security	Basic defaults	Fully customized by you
Subnets	Pre-created by AWS	You design every subnet
Best for	Beginners, experiments	Companies, real projects

When you launched your EC2 instance in Chapter 8 practical — it automatically went into the DEFAULT VPC without you realizing it!

🔗 **VPC = Your own private, isolated network inside AWS — where you control the boundaries, the roads, the gates, and the security of everything running inside it.**

CHAPTER 10 — How VPC Components Interact & Why You Need VPC

How All Components Interact — Full Traffic Flow

Let us trace what happens when a user opens YOUR website hosted on AWS:

Step 1: User opens website from their phone



Step 2: Request hits the INTERNET



Step 3: INTERNET GATEWAY — the main gate allows the request in



Step 4: NACL CHECK — subnet-level guard checks: is this traffic allowed?



Step 5: Enters PUBLIC SUBNET — the public zone of your VPC



Step 6: ROUTE TABLE — where should this go? Routes to EC2 web server



Step 7: SECURITY GROUP CHECK — is Port 80 open? Yes. Let it through.



Step 8: EC2 WEB SERVER processes the request — needs data from DB



Step 9: Talks to PRIVATE SUBNET — fully internal, no internet involved



Step 10: RDS DATABASE returns data to web server



Step 11: Response travels back the same path to the user

Real World Example — How a Fintech App Uses VPC

VPC: 10.0.0.0/16

Availability Zone A (Mumbai)

Public Subnet 10.0.1.0/24 → Load Balancer + Web Server (EC2)

Private Subnet 10.0.2.0/24 → App Server + RDS Database

Availability Zone B (Mumbai backup)

Public Subnet 10.0.3.0/24 → Load Balancer + Web Server (EC2)
Private Subnet 10.0.4.0/24 → App Server + RDS Replica

Internet Gateway → Public internet access
NAT Gateway → Private servers can download updates
Route Tables → Traffic directed correctly to each layer

Result: Users reach website through Load Balancer. App servers process transactions privately. Database NEVER exposed to internet. If one AZ goes down — other AZ keeps everything running. This is how SBI, HDFC and every serious fintech company structures their cloud.

? Why Do You Need VPC?

Reason 1 — Security and Isolation

Without VPC all your AWS resources are on a shared public network. Anyone could potentially reach your database. VPC creates a completely isolated private network that only you control.

Reason 2 — You Control Your Network

Without VPC: AWS decides your network rules.
With VPC: YOU decide your own IP ranges, subnets, routing rules, firewall rules, and access policies.

Reason 3 — Compliance and Regulations

Banks, hospitals, and government organizations are legally required to keep sensitive data in isolated networks. VPC is how they meet GDPR, PCI-DSS, and HIPAA compliance on AWS.

Reason 4 — Multi-Layer Security

Layer 1: NACL — blocks at subnet level
Layer 2: Security Group — blocks at instance level
Layer 3: IAM — controls who can manage resources

Three layers = an attacker has to break through all three.

Reason 5 — Production Grade Architecture

Every real production application on AWS uses VPC. Without it you cannot build:

- High availability across multiple Availability Zones
- Proper separation of web, app, and database tiers
- Private internal communication between microservices
- Secure connections that meet enterprise standards

VPC Connects Everything You Have Learned

What You Learned	How VPC Connects to It
EC2 (Chapter 7)	Every EC2 instance runs inside a subnet inside a VPC
S3 (Chapter 1)	Connected privately via VPC Endpoint — no internet needed
IAM (Chapter 5)	Controls WHO can manage and access resources inside VPC
Security Groups (Chapter 8)	The instance-level firewall layer inside VPC
Jenkins on EC2	Your Jenkins server lives inside the Default VPC right now

Quick Revision — All 7 Components in One Table

Component	Job	Analogy
Subnet	Divide VPC into public and private zones	Rooms in a house
Internet Gateway	Connect VPC to public internet	Main front door
Route Table	Direct traffic to the right destination	Road signboards
Security Group	Firewall at instance level — allow rules only	Room door lock
NACL	Firewall at subnet level — allow and deny rules	Floor security guard
NAT Gateway	Private servers reach internet — one way only	One-way mirror
VPC Peering	Private connection between two VPCs	Private tunnel to neighbour

 **VPC components are like the complete infrastructure of a secure private building — each piece has a job, and together they make sure the right traffic reaches the right place and everything else is blocked.**

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